

INTRODUCTION TO WEB PROGRAMMING

PRACTICAL FILE

RAMANUJAN COLLEGE



UNIVERSITY OF DELHI

DSE: INTRODUCTION TO WEB PROGRAMMING

SEMESTER – 4

SESSION – 2025-26

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Seminar			
Day	Schedule		Topic
	Start Time	End Time	
Monday	8:00 AM	5:00 PM	Introduction to HTML
			Important HTML Tags
Tuesday	8:00 AM	11:00 AM	CSS-1
	11:00 AM	2:00 PM	
	2:00 PM	5:00 PM	JavaScript-1
Wednesday	8:00 AM	12 Noon	JavaScript-2
Thursday	8:00 AM	12 Noon	JQuery
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HTML

PRACTICAL – 1

Ques: Create an HTML document with following formatting – Bold, Italics, Underline, Colours, Headings, Title, Font and Font Width, Background, Paragraph, Line Breaks, Horizontal Line, marquee text.

Input:-

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>HTML Formatting Demo</title>
5  </head>
6
7  <body style="background-color: ■lightyellow;">
8
9      <!-- Heading -->
10     <h1 style="color: ■darkblue;">Welcome to HTML Formatting</h1>
11     <h2>This is a Subheading</h2>
12     <h3>Smaller Heading</h3>
13
14     <!-- Paragraph with formatting -->
15     <p style="font-family: Arial; font-size: 18px; color: ■black;">
16         This is a paragraph demonstrating different text styles.<br>
17         <b>This text is bold.</b><br>
18         <i>This text is italic.</i><br>
19         <u>This text is underlined.</u><br>
20         <font color="red" size="4">This text is colored and resized.</font>
21     </p>
22
23     <!-- Line break -->
24     This is line one.<br>
25     This is line two after a line break.<br><br>
26
27     <!-- Horizontal line -->
28     <hr>
29
30     <!-- Another paragraph -->
31     <p style="font-family: 'Courier New'; font-size: 16px; color: ■green;">
32         This paragraph uses a different font style and color.
33     </p>
```

```
35     <!-- Marquee text -->
36     <marquee behavior="scroll" direction="left" style="color: purple; font-size: 20px;">
37         This is scrolling marquee text!
38     </marquee>
39
40 </body>
41 </html>
```

Output:-

Welcome to HTML Formatting

This is a Subheading

Smaller Heading

This is a paragraph demonstrating different text styles.

This text is bold.

This text is italic.

This text is underlined.

This text is colored and resized.

This is line one.

This is line two after a line break.

This paragraph uses a different font style and color.

This is scrolling marquee text!

PRACTICAL – 2

Ques: Create an HTML document with Ordered and Unordered lists.

Input:-

```
<!DOCTYPE html>
<html>
<head>
  <title>Ordered and Unordered Lists</title>
</head>

<body>

  <h1>HTML Lists Example</h1>

  <!-- Ordered List -->
  <h2>Ordered List</h2>
  <ol>
    <li>Wake up</li>
    <li>Brush teeth</li>
    <li>Have breakfast</li>
    <li>Go to college</li>
  </ol>

  <!-- Unordered List -->
  <h2>Unordered List</h2>
  <ul>
    <li>Apple</li>
    <li>Banana</li>
    <li>Mango</li>
    <li>Orange</li>
  </ul>

</body>
</html>
```

Output:-

HTML Lists Example

Ordered List

1. Wake up
2. Brush teeth
3. Have breakfast
4. Go to college

Unordered List

- Apple
- Banana
- Mango
- Orange

PRACTICAL – 3

Ques: Create an HTML document demonstrating use of images in webpages (including images as logos, cell data in a table, background of a table, clickable icons, etc.)

Input:-

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>Images in Webpages</title>
5  </head>
6
7  <body style = "font-family: Arial;">
8
9    <h1> Wesbite with images </h1>
10   <img src = "download.jpg" alt = "Website Logo" width = "150" height = "100">
11
12   <hr>
13
14   <h2> Clickable Icon </h2>
15   <a href = "https://www.google.com" target="_blank">
16     <img src = "google icon.jpg" alt="Google Icon" width="80" height="80">
17   </a>
18
19   <hr>
20
21   <h2>Images Inside Table</h2>
22
23   <table border="1" cellpadding="10" background = "light blue.jpg">
24
25     <tr>
26       <th>Student</th>
27       <th>Photo</th>
28       <th>Course</th>
29     </tr>
```

```
30
31     <tr>
32         <td>Amy</td>
33         <td>
34             
35         </td>
36         <td>BCA</td>
37     </tr>
38
39     <tr>
40         <th>Bibi</th>
41         <td>
42             
43         </td>
44         <td>B.Des</td>
45     </tr>
46
47 </table>
48
49 <hr>
50
```

```
50
51 <h2>Background Image Example</h2>
52
53 <div style="width: 500px; height: 200 px;
54     background-image: url('https://images.unsplash.com/photo-1506744038136-46273834b3fb');
55     background-size: cover; color: white; padding: 20px;">
56
57     <h3>Text on Background Image</h3>
58     <p>This section uses an image as background.</p>
59
60 </div>
61
62 </body>
63 </html>
```

Output:-

Wesbite with images



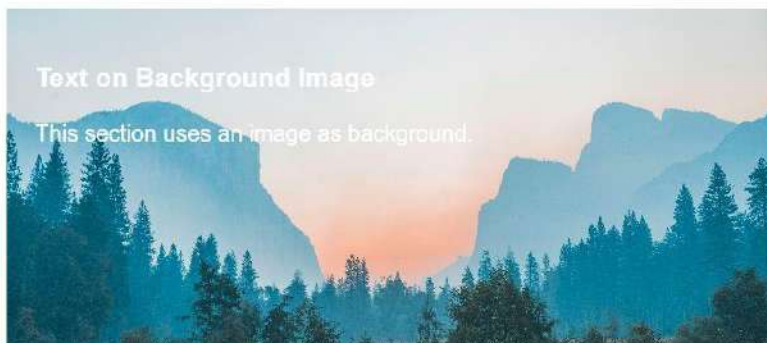
Clickable Icon



Images Inside Table

Student	Photo	Course
Amy		BCA
Bibi		B.Des

Background Image Example



PRACTICAL – 4

Ques: Create an HTML document to demonstrate Internal and External linking.

Input:-

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Internal and External Linking</title>
5  </head>
6
7  <body>
8
9      <h1>HTML Linking Example</h1>
10
11      <!-- Internal Links -->
12      <h2>Internal Linking</h2>
13
14      <a href="#about">Go to About Section</a><br>
15      <a href="#contact">Go to Contact Section</a>
16
17      <hr>
18
19      <!-- External Links -->
20      <h2>External Linking</h2>
21
22      <a href="https://www.google.com" target="_blank">
23          Visit Google
24      </a><br><br>
25
26      <a href="https://www.wikipedia.org" target="_blank">
27          Visit Wikipedia
28      </a>
29
30      <hr>
31
32      <!-- About Section -->
33      <h2 id="about">About Section</h2>
34
```



```
35     <p>
36         This is the About section of the webpage.
37         Internal links help users move to different parts
38         of the same page.
39     </p>
40
41     <hr>
42
43     <!-- Contact Section -->
44     <h2 id="contact">Contact Section</h2>
45
46     <p>
47         This is the Contact section of the webpage.
48     </p>
49
50 </body>
51 </html>
```

Output:-

HTML Linking Example

Internal Linking

[Go to About Section](#)

[Go to Contact Section](#)

External Linking

[Visit Google](#)

[Visit Wikipedia](#)

About Section

This is the About section of the webpage. Internal links help users move to different parts of the same page.

Contact Section

This is the Contact section of the webpage.

PRACTICAL – 5

Ques: Create an HTML to display the following table:

Table Example

Seminar			
Day	Schedule		Topic
	Start Time	End Time	
Monday	8:00 AM	5:00 PM	Introduction to HTML
			Important HTML Tags
Tuesday	8:00 AM	11:00 AM	CSS-1
	11:00 AM	2:00 PM	
	2:00 PM	5:00 PM	JavaScript-1
Wednesday	8:00 AM	12 Noon	JavaScript-2
Thursday	8:00 AM	12 Noon	JQuery
	1:00 PM	3:00 PM	Valedictory

Input:-

```

1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Table Example</title>
5
6      <style>
7          body{
8              background-color: #dcdcdc;
9              font-family: "Times New Roman";
10         }
11
12         h1{
13             text-align: center;
14             color: blue;
15         }
16
17         hr{
18             border: 3px solid brown;
19             width: 95%;
20         }
21
22         .outer{
23             width: 760px; /* made broader */
24             margin: auto;
25             border: 6px solid gray;
26             padding: 12px;
27             background: #dcdcdc;
28         }

```

```
30     table{
31         width:100%;
32         border-collapse:separate;
33         border-spacing:4px;
34     }
35
36     th, td{
37         border:2px solid blue;
38         padding:12px;
39         font-size:22px;
40         background: #dcdcdc;
41     }
42
43     th{
44         background: yellow;
45     }
46
47     .center{
48         text-align:center;
49     }
50
51     .topic{
52         background: #eeeeee;
53     }
54 </style>
55 </head>
56
57 <body>
58
59 <h1>Table Example</h1>
60
```

```
61 <hr>
62
63 <div class="outer">
64
65 <table>
66
67     <tr>
68         <th colspan="4" class="center">Seminar</th>
69     </tr>
70
71     <tr>
72         <th rowspan="2" class="center">Day</th>
73         <th colspan="2" class="center">Schedule</th>
74         <th rowspan="2" class="center">Topic</th>
75     </tr>
76
77     <tr>
78         <th class="center">Start Time</th>
79         <th class="center">End Time</th>
80     </tr>
81
82     <tr>
83         <td rowspan="2">Monday</td>
84         <td rowspan="2">8:00 AM</td>
85         <td rowspan="2">5:00 PM</td>
86         <td class="topic">Introduction to HTML</td>
87     </tr>
88
89     <tr>
90         <td class="topic">Important HTML Tags</td>
91     </tr>
```

```
93     <tr>
94         <td rowspan="3">Tuesday</td>
95         <td>8:00 AM</td>
96         <td>11:00 AM</td>
97         <td rowspan="2">CSS-1</td>
98     </tr>
99
100    <tr>
101        <td>11:00 AM</td>
102        <td>2:00 PM</td>
103    </tr>
104
105    <tr>
106        <td>2:00 PM</td>
107        <td>5:00 PM</td>
108        <td>JavaScript-1</td>
109    </tr>
110
111    <tr>
112        <td>Wednesday</td>
113        <td>8:00 AM</td>
114        <td>12 Noon</td>
115        <td>JavaScript-2</td>
116    </tr>
117
118    <tr>
119        <td rowspan="2">Thursday</td>
120        <td>8:00 AM</td>
121        <td>12 Noon</td>
122        <td>jQuery</td>
123    </tr>
```

```
125     <tr>
126         <td>1:00 PM</td>
127         <td>3:00 PM</td>
128         <td>Valedictory</td>
129     </tr>
130
131 </table>
132
133 </div>
134
135 </body>
136 </html>
```

Output:-

Seminar			
Day	Schedule		Topic
	Start Time	End Time	
Monday	8:00 AM	5:00 PM	Introduction to HTML
			Important HTML Tags
Tuesday	8:00 AM	11:00 AM	CSS-1
	11:00 AM	2:00 PM	
	2:00 PM	5:00 PM	JavaScript-1
Wednesday	8:00 AM	12 Noon	JavaScript-2
Thursday	8:00 AM	12 Noon	JQuery
	1:00 PM	3:00 PM	Valedictory

PRACTICAL – 6

Ques: Create an HTML page demonstrating the following semantic tags:

- a. Header
- b. Nav
- c. Main
- d. Section
- e. Footer
- f. Details
- g. Summary
- h. Figure
- i. Figure caption

Input:-

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Semantic Tags Example</title>
5
6      <style>
7          body{
8              font-family: Arial, sans-serif;
9              margin: 0;
10             background-color: #f2f2f2;
11         }
12
13         header{
14             background-color: darkblue;
15             color: white;
16             text-align: center;
17             padding: 20px;
18         }
19
20         nav{
21             background-color: #333;
22             padding: 10px;
23             text-align: center;
24         }
25
```

```
26     nav a{
27         color: ■white;
28         text-decoration: none;
29         margin: 15px;
30     }
31
32     main{
33         padding: 20px;
34     }
35
36     section{
37         background-color: ■white;
38         padding: 15px;
39         margin-bottom: 20px;
40         border-radius: 5px;
41     }
42
43     footer{
44         background-color: ■darkblue;
45         color: ■white;
46         text-align: center;
47         padding: 15px;
48     }
49
50     figure{
51         text-align: center;
52     }
53
54     figcaption{
55         font-style: italic;
56         margin-top: 5px;
57     }
```

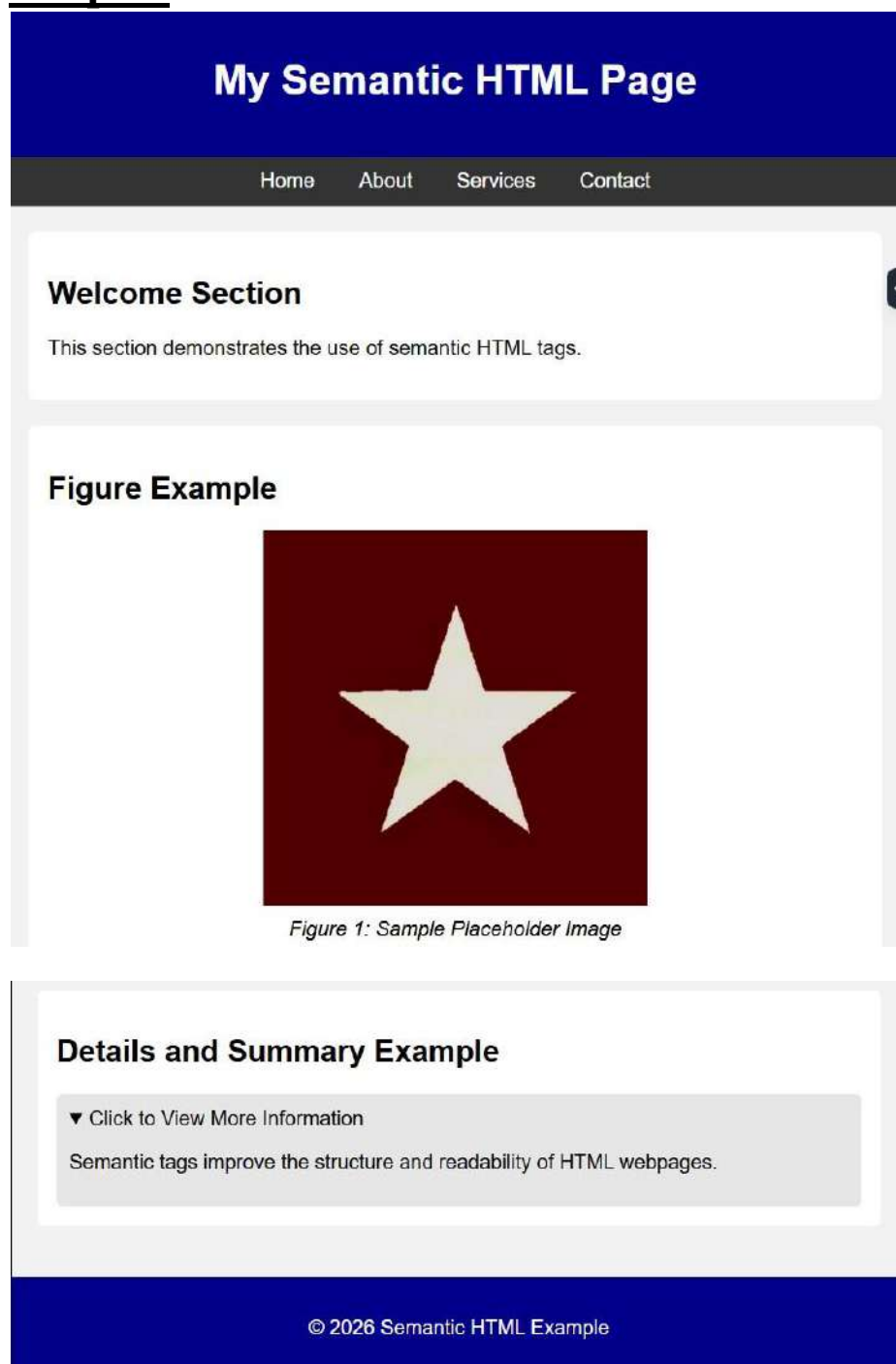
```
59     details{
60         background-color: #e6e6e6;
61         padding: 10px;
62         border-radius: 5px;
63     }
64 </style>
65 </head>
66
67 <body>
68
69     <!-- Header -->
70     <header>
71         <h1>My Semantic HTML Page</h1>
72     </header>
73
74     <!-- Navigation -->
75     <nav>
76         <a href="#">Home</a>
77         <a href="#">About</a>
78         <a href="#">Services</a>
79         <a href="#">Contact</a>
80     </nav>
81
82     <!-- Main Content -->
83     <main>
84
85         <!-- Section -->
86         <section>
87             <h2>Welcome Section</h2>
88             <p>
89                 This section demonstrates the use of semantic HTML tags.
90             </p>
91         </section>
```



```
93     <!-- Figure and Figcaption -->
94     <section>
95         <h2>Figure Example</h2>
96
97         <figure>
98             
101
102             <figcaption>
103                 Figure 1: Sample Placeholder Image
104             </figcaption>
105         </figure>
106     </section>
107
108     <!-- Details and Summary -->
109     <section>
110         <h2>Details and Summary Example</h2>
111
112         <details>
113             <summary>Click to View More Information</summary>
114
115             <p>
116                 Semantic tags improve the structure and readability
117                 of HTML webpages.
118             </p>
119         </details>
120     </section>
121
122 </main>
123
```

```
124     <!-- Footer -->
125     <footer>
126         <p>© 2026 Semantic HTML Example</p>
127     </footer>
128
129 </body>
130 </html>
```

Output:



PRACTICAL – 7

Ques: Create a student registration form using HTML which has the following controls: a. Text Box

b. Text Area

c. Dropdown box

d. Option/radio buttons

e. Check boxes

f. Reset and Submit button

On pressing the Submit button, a message “Form submitted” should be displayed.

Input:-

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Student Registration Form</title>
5
6      <script>
7          function showMessage() {
8              alert("Form submitted");
9          }
10     </script>
11
12     <style>
13         body{
14             font-family: Arial, sans-serif;
15             background-color: #f2f2f2;
16         }
17
18         .form-box{
19             width: 450px;
20             margin: auto;
21             background: white;
22             padding: 20px;
23             border-radius: 8px;
24         }
```

```
26     h2{
27         text-align: center;
28         color: darkblue;
29     }
30
31     label{
32         font-weight: bold;
33     }
34
35     input, textarea, select{
36         width: 100%;
37         padding: 8px;
38         margin-top: 5px;
39         margin-bottom: 15px;
40     }
41
42     .inline{
43         width: auto;
44     }
45
46     .buttons{
47         text-align: center;
48     }
49
50     button{
51         padding: 10px 20px;
52         margin: 10px;
53     }
54 </style>
55 </head>
```

```
57 <body>
58
59 <div class="form-box">
60
61     <h2>Student Registration Form</h2>
62
63     <form onsubmit="showMessage()">
64
65         <!-- Text Box -->
66         <label>Student Name:</label>
67         <input type="text" placeholder="Enter your name">
68
69         <!-- Text Area -->
70         <label>Address:</label>
71         <textarea rows="4" placeholder="Enter your address"></textarea>
72
73         <!-- Dropdown Box -->
74         <label>Course:</label>
75         <select>
76             <option>B.Sc</option>
77             <option>BCA</option>
78             <option>B.Tech</option>
79             <option>BBA</option>
80             <option>MCA</option>
81         </select>
82
83         <!-- Radio Buttons -->
84         <label>Gender:</label><br>
85
86         <input type="radio" name="gender" class="inline"> Male
87         <input type="radio" name="gender" class="inline"> Female
88         <input type="radio" name="gender" class="inline"> Other
89
90         <br><br>
91
92         <!-- Check Boxes -->
93         <label>Hobbies:</label><br>
94
95         <input type="checkbox" class="inline"> Reading
96         <input type="checkbox" class="inline"> Sports
97         <input type="checkbox" class="inline"> Music
98
99         <br><br>
100
101         <!-- Buttons -->
102         <div class="buttons">
103             <button type="submit">Submit</button>
104             <button type="reset">Reset</button>
105         </div>
106     </form>
107
108 </div>
109
110 </body>
111 </html>
```

Output:-

Student Registration Form

Student Name:

Address:

Course:

Gender:

☐ Male ☐ Female ☐ Other

Hobbies:

☐ Reading ☐ Sports ☐ Music

This page says

Form submitted

Address:

Course:

Gender:

☐ Male ☒ Female ☐ Other

Hobbies:

☒ Reading ☐ Sports ☒ Music

Input:-

```

1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Computer Science Department</title>
5
6      <!-- External CSS Link -->
7      <link rel="stylesheet" href="style1.css">
8  </head>
9
10 <body>
11
12 <!-- Header -->
13 <div id="header">
14     <h1>Computer Science Department</h1>
15     <p>Welcome to Our Department Website</p>
16 </div>
17
18 <!-- Navigation Menu -->
19 <nav>
20     <ul>
21
22         <li>
23             <a href="#">Home</a>
24         </li>
25
26         <li>
27             <a href="#">Faculty</a>
28
29             <ul>
30                 <li><a href="#">Professors</a></li>
31                 <li><a href="#">Assistant Professors</a></li>
32                 <li><a href="#">Lab Staff</a></li>
33             </ul>
34         </li>

```

```
36         <li>
37             <a href="#">Courses</a>
38
39             <ul>
40                 <li><a href="#">BCA</a></li>
41                 <li><a href="#">B.Tech</a></li>
42                 <li><a href="#">MCA</a></li>
43             </ul>
44         </li>
45
46         <li>
47             <a href="#">Activities</a>
48
49             <ul>
50                 <li><a href="#">Workshops</a></li>
51                 <li><a href="#">Hackathons</a></li>
52                 <li><a href="#">Seminars</a></li>
53             </ul>
54         </li>
55
56         <li>
57             <a href="#">Contact</a>
58         </li>
59     </ul>
60 </nav>
61
62 <!-- Main Content -->
63 <div class="content">
64
65
```

```
66     <div class="section-box">
67         <h2>About the Department</h2>
68
69         <p>
70             The Computer Science Department provides quality education
71             in programming, networking, database systems, and web technologies.
72         </p>
73     </div>
74
75     <div class="section-box">
76         <h2>Latest Activities</h2>
77
78         <p>
79             Students recently participated in coding competitions,
80             technical workshops, and innovation projects.
81         </p>
82     </div>
83
84 </div>
85
86 <!-- Footer -->
87 <footer>
88     <p>© 2026 Computer Science Department</p>
89 </footer>
90
91 </body>
92 </html>
```

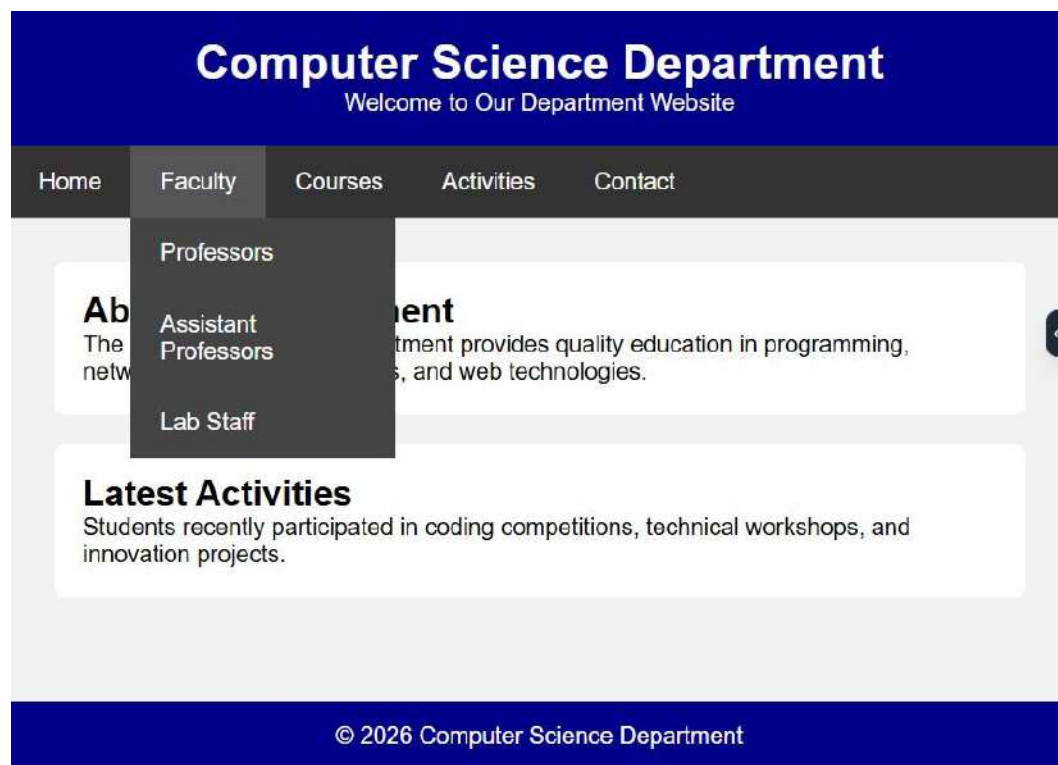
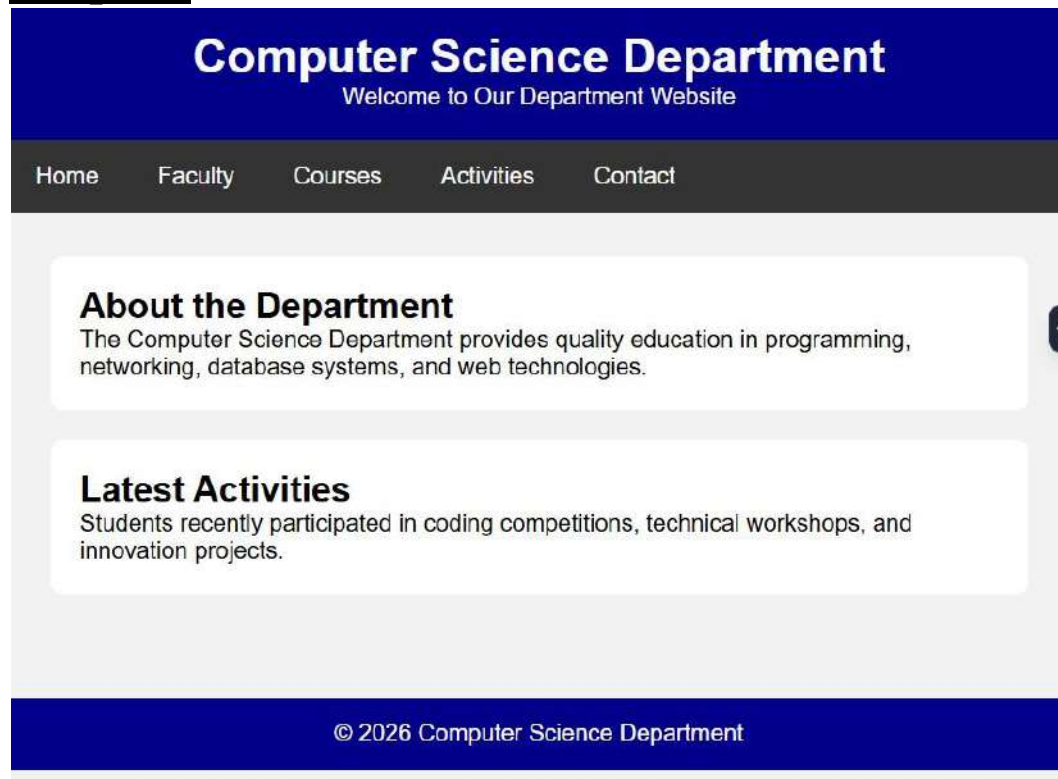


```
1  /* Universal Selector */
2  *{
3      margin: 0;
4      padding: 0;
5      box-sizing: border-box;
6  }
7
8  /* Body Style */
9  body{
10     font-family: Arial, sans-serif;
11     background-color: #f2f2f2;
12 }
13
14 /* Header using ID Selector */
15 #header{
16     background-color: darkblue;
17     color: white;
18     text-align: center;
19     padding: 20px;
20 }
21
22 /* Navigation Bar */
23 nav{
24     background-color: #333;
25 }
26
27 nav ul{
28     list-style: none;
29     display: flex;
30 }
31
32 nav ul li{
33     position: relative;
34 }
```

```
35
36     nav ul li a{
37         display: block;
38         color: white;
39         padding: 15px 20px;
40         text-decoration: none;
41     }
42
43     /* Hover Effect */
44     nav ul li a:hover{
45         background-color: #555;
46     }
47
48     /* Dropdown Menu */
49     nav ul li ul{
50         display: none;
51         position: absolute;
52         background-color: #444;
53         min-width: 180px;
54     }
55
56     nav ul li:hover ul{
57         display: block;
58     }
59
60     nav ul li ul li{
61         width: 100%;
62     }
63
64     /* Class Selector */
65     .content{
66         padding: 30px;
67     }
68
```

```
69     .section-box{
70         background-color: white;
71         padding: 20px;
72         margin-bottom: 20px;
73         border-radius: 8px;
74     }
75
76     /* Footer */
77     footer{
78         background-color: darkblue;
79         color: white;
80         text-align: center;
81         padding: 15px;
82         margin-top: 20px;
83     }
```

Output:-



JAVASCRIPT

PRACTICAL – 1

Ques: Write a JavaScript code to accept a number from the user through the prompt and print its multiplication table.

Input:-

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Multiplication Table</title>
5  </head>
6
7  <body>
8
9      <h2>Multiplication Table using JavaScript</h2>
10
11     <script>
12
13         // Accept number from user
14         let num = prompt("Enter a number:");
15
16         document.write("<h3>Multiplication Table of " + num + "</h3>");
17
18         // Loop to print table
19         for(let i = 1; i <= 10; i++)
20         {
21             document.write(num + " x " + i + " = " + (num * i) + "<br>");
22         }
23
24     </script>
25
26 </body>
27 </html>
```

Output:-

This page says

Enter a number:

OK Cancel

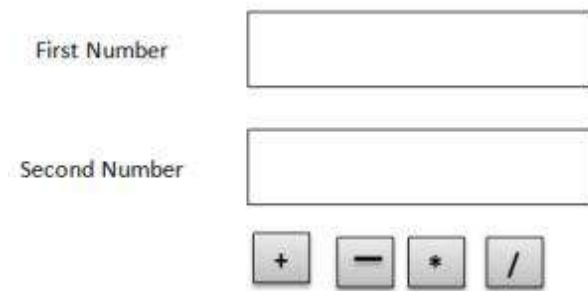
Multiplication Table using JavaScript

Multiplication Table of 754

$754 \times 1 = 754$
 $754 \times 2 = 1508$
 $754 \times 3 = 2262$
 $754 \times 4 = 3016$
 $754 \times 5 = 3770$
 $754 \times 6 = 4524$
 $754 \times 7 = 5278$
 $754 \times 8 = 6032$
 $754 \times 9 = 6786$
 $754 \times 10 = 7540$

PRACTICAL – 2

Ques: Create a HTML form having two text boxes and 4 buttons (as shown below) .



The image shows a simple HTML form. It consists of two text input boxes. The first box is labeled "First Number" and the second box is labeled "Second Number". Below the second box, there are four buttons: a plus sign (+), a minus sign (-), a multiplication sign (*), and a division sign (/).

Input:-

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Simple Calculator</title>
5
6      <style>
7
8          body{
9              font-family: Arial, sans-serif;
10             background-color: #f2f2f2;
11         }
12
13         .container{
14             width: 500px;
15             margin: 30px auto;
16         }
17
18         .row{
19             margin-bottom: 25px;
20         }
21
22         label{
23             display: inline-block;
24             width: 170px;
25             font-size: 20px;
26         }
27
```

```
28     input{
29         width: 230px;
30         height: 40px;
31         font-size: 18px;
32     }
33
34     .buttons{
35         margin-left: 175px;
36     }
37
38     button{
39         width: 45px;
40         height: 45px;
41         margin-right: 12px;
42         font-size: 24px;
43         font-weight: bold;
44         cursor: pointer;
45     }
46
47 </style>
48
49 <script>
50
51     function calculate(operator)
52     {
53         let num1 = parseFloat(document.getElementById("num1").value);
54
55         let num2 = parseFloat(document.getElementById("num2").value);
56
57         let result;
```



```
59         if(operator == '+')
60         {
61             result = num1 + num2;
62         }
63
64         else if(operator == '-')
65         {
66             result = num1 - num2;
67         }
68
69         else if(operator == '*')
70         {
71             result = num1 * num2;
72         }
73
74         else if(operator == '/')
75         {
76             result = num1 / num2;
77         }
78
79         alert("Result = " + result);
80     }
81
82     </script>
83
84 </head>
85
86 <body>
87
88     <div class="container">
89
```



```
90     <div class="row">
91         <label>First Number</label>
92
93         <input type="text" id="num1">
94     </div>
95
96     <div class="row">
97         <label>Second Number</label>
98
99         <input type="text" id="num2">
100    </div>
101
102    <div class="buttons">
103
104        <button onclick="calculate('+')">+</button>
105
106        <button onclick="calculate('-')">-</button>
107
108        <button onclick="calculate('*')">*</button>
109
110        <button onclick="calculate('/')">/</button>
111
112    </div>
113
114    </div>
115
116    </body>
117    </html>
```

Output:-

First Number	84854
Second Number	62305
	+ - * /

First Number	84854
Second Number	62305
	+ - * /

This page says

Result = 22549

OK

PRACTICAL – 3

Ques: Write a JavaScript code to change the background colour of a text box when text box gets focus.

Input:-

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Textbox Focus Example</title>
5
6      <style>
7
8          body{
9              font-family: Arial, sans-serif;
10             margin: 40px;
11         }
12
13         input{
14             width: 250px;
15             height: 35px;
16             font-size: 18px;
17             padding-left: 10px;
18         }
19
20     </style>
21
22     <script>
23
24         function changeColor(element)
25         {
26             element.style.backgroundColor = "lightblue";
27         }
28
29         function resetColor(element)
30         {
31             element.style.backgroundColor = "white";
32         }
33
34     </script>
```

```
35
36 </head>
37
38 <body>
39
40   <h2>Focus Event Example</h2>
41
42   <input type="text"
43         placeholder="Click inside the textbox"
44
45         onfocus="changeColor(this)"
46
47         onblur="resetColor(this)">
48
49 </body>
50 </html>
```

Output:-

Focus Event Example

Focus Event Example

PRACTICAL – 4

Ques: Write a JavaScript code for Q7 (of HTML above) to input and validate all data. Create functions to perform validation of each element, for example:

- a. Roll number is a 7-digit numeric value.
- b. Name should be an alphabetical value (String).
- c. Non-empty fields like DOB.

Input:-

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Student Registration Form Validation</title>
5
6      <style>
7
8          body{
9              font-family: Arial, sans-serif;
10             background-color: #f2f2f2;
11         }
12
13         .form-box{
14             width: 450px;
15             margin: auto;
16             background: white;
17             padding: 20px;
18             border-radius: 8px;
19         }
20
21         h2{
22             text-align: center;
23             color: darkblue;
24         }
25
26         label{
27             font-weight: bold;
28         }
29
30         input, textarea, select{
31             width: 100%;
32             padding: 8px;
33             margin-top: 5px;
34             margin-bottom: 15px;
35         }
```

```
37     .inline{
38         width: auto;
39     }
40
41     .buttons{
42         text-align: center;
43     }
44
45     button{
46         padding: 10px 20px;
47         margin: 10px;
48     }
49
50 </style>
51
52 <script>
53
54     // Validate Roll Number
55     function validateRoll()
56     {
57         let roll = document.getElementById("roll").value;
58
59         let pattern = /^[0-9]{7}$/;
60
61         if(!pattern.test(roll))
62         {
63             alert("Roll Number must be a 7-digit numeric value");
64
65             return false;
66         }
67
68         return true;
69     }
```

```
71      // Validate Name
72      function validateName()
73      {
74          let name = document.getElementById("name").value;
75
76          let pattern = /^[A-Za-z ]+$/;
77
78          if(!pattern.test(name))
79          {
80              alert("Name should contain only alphabets");
81
82              return false;
83          }
84
85          return true;
86      }
87
88      // Validate DOB
89      function validateDOB()
90      {
91          let dob = document.getElementById("dob").value;
92
93          if(dob == "")
94          {
95              alert("DOB field cannot be empty");
96
97              return false;
98          }
99
100         return true;
101     }
```

```
103         // Main Validation Function
104         function validateForm()
105         {
106             if(validateRoll() &&
107                validateName() &&
108                validateDOB())
109             {
110                 alert("Form submitted");
111                 return true;
112             }
113             return false;
114         }
115     }
116     </script>
117 </head>
118 <body>
119     <div class="form-box">
120         <h2>Student Registration Form</h2>
121         <form onsubmit="return validateForm()">
122             <!-- Roll Number -->
123             <label>Roll Number:</label>
124             <input type="text" id="roll">
125         </form>
126     </div>
127 </body>
128 </html>
```



```
135      <!-- Name -->
136      <label>Student Name:</label>
137
138      <input type="text" id="name">
139
140      <!-- DOB -->
141      <label>Date of Birth:</label>
142
143      <input type="date" id="dob">
144
145      <!-- Address -->
146      <label>Address:</label>
147
148      <textarea rows="4"></textarea>
149
150      <!-- Dropdown -->
151      <label>Course:</label>
152
153      <select>
154          <option>B.Sc</option>
155          <option>BCA</option>
156          <option>B.Tech</option>
157          <option>BBA</option>
158          <option>MCA</option>
159      </select>
160
161      <!-- Radio Buttons -->
162      <label>Gender:</label><br><br>
163
164      <input type="radio" name="gender" class="inline"> Male
165
166      <input type="radio" name="gender" class="inline"> Female
```



```
168         <input type="radio" name="gender" class="inline"> Other
169
170     <br><br>
171
172     <!-- Checkboxes -->
173     <label>Hobbies:</label><br><br>
174
175     <input type="checkbox" class="inline"> Reading
176
177     <input type="checkbox" class="inline"> Sports
178
179     <input type="checkbox" class="inline"> Music
180
181     <br><br>
182
183     <!-- Buttons -->
184     <div class="buttons">
185
186         <button type="submit">Submit</button>
187
188         <button type="reset">Reset</button>
189
190     </div>
191
192 </form>
193
194 </div>
195
196 </body>
197 </html>
```

Output:-

This page says

Roll Number must be a 7-digit numeric value

OK

56800632

Student Name:

Akansha Mehrotra

Date of Birth:

01-03-2006

Address:

Paradise Village, Phase 2

Course:

BBA

Gender:

☐ Male ☒ Female ☐ Other

Hobbies:

☒ Reading ☐ Sports ☒ Music

Submit

Reset

This page says

Name should contain only alphabets

OK

5670632

Student Name:

Khy@t1 Sh@rm@

Date of Birth:

15-08-2005

**Address:**

Sunny Fields, Phase 5

Course:

BCA

**Gender:**☐ Male ☒ Female ☐ Other**Hobbies:**☒ Reading ☒ Sports ☒ Music

Submit

Reset

This page says

DOB field cannot be empty

OK

5670632

Student Name:

Veronica Kapoor

Date of Birth:

dd-mm-yyyy

**Address:**

Retreat Drive, Sector 28

Course:

MCA

Gender:☐ Male ☒ Female ☐ Other**Hobbies:**☒ Reading ☒ Sports ☒ Music

Submit

Reset

This page says

Form submitted

OK

5493046

Student Name:

Ahaan Kanojia

Date of Birth:

27 - 11 - 2005



Address:

The Verandas, Sector - 23

Course:

B.Tech



Gender:

☒ Male ☐ Female ☐ Other

Hobbies:

☒ Reading ☒ Sports ☒ Music

Submit

Reset

JQUERY AND JSON

PRACTICAL – 1

Ques: Change text colour and contents using button click events using jQuery.

Input:-

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>jQuery Text Change Example</title>
5
6      <!-- jQuery Library -->
7      <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
8
9      <style>
10
11          body{
12              font-family: Arial, sans-serif;
13              text-align: center;
14              margin-top: 50px;
15          }
16
17          #message{
18              font-size: 28px;
19              color: ☐ black;
20              margin-bottom: 20px;
21          }
22
23          button{
24              padding: 10px 20px;
25              font-size: 18px;
26              margin: 10px;
27              cursor: pointer;
28          }
```

```
30     </style>
31
32     <script>
33
34         $(document).ready(function(){
35
36             // Change text color
37             $("#colorBtn").click(function(){
38
39                 $("#message").css("color", "blue");
40
41             });
42
43             // Change text content
44             $("#contentBtn").click(function(){
45
46                 $("#message").text("Text has been changed using jQuery!");
47
48             });
49
50         });
51     </script>
52
53 </head>
54
55 <body>
56
57     <h2 id="message">
58         Welcome to jQuery Example
59     </h2>
```

```
62     <button id="colorBtn">
63         Change Color
64     </button>
65
66     <button id="contentBtn">
67         Change Text
68     </button>
69
70 </body>
71 </html>
```

Output:-

Welcome to jQuery Example

Change Color

Change Text

Welcome to jQuery Example

Change Color

Change Text

Text has been changed using jQuery!

Change Color

Change Text

PRACTICAL – 2

Ques: Select elements using ID, class, elements name, attribute name.

Input:-

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>jQuery Selectors Example</title>
5
6      <!-- jQuery Library -->
7      <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
8
9      <style>
10
11          body{
12              font-family: Arial, sans-serif;
13              margin: 30px;
14          }
15
16          .box{
17              padding: 10px;
18              margin: 10px 0;
19              border: 2px solid gray;
20          }
21
22      </style>
23
24      <script>
25
26          $(document).ready(function(){
27
28              // JSON Data
29              let student = {
30                  "name" : "Aman",
31                  "course" : "BCA",
32                  "city" : "Delhi"
33              };

```

```
35         // ID Selector
36         $("#idBtn").click(function(){
37
38             $("#heading").css("color", "blue");
39
40         });
41
42         // Class Selector
43         $("#classBtn").click(function(){
44
45             $(".box").css("background-color", "lightyellow");
46
47         });
48
49         // Element Selector
50         $("#elementBtn").click(function(){
51
52             $("p").css("font-size", "22px");
53
54         });
55
56         // Attribute Selector
57         $("#attrBtn").click(function(){
58
59             $("input[type='text']").css("border",
60             "3px solid red");
61
62         });
63
64         // Display JSON Data
```

```
65         $("#jsonBtn").click(function(){
66
67             $("#jsonData").html(
68                 "Name: " + student.name + "<br>" +
69                 "Course: " + student.course + "<br>" +
70                 "City: " + student.city
71             );
72
73         });
74
75     });
76
77     </script>
78
79 </head>
80
81 <body>
82
83     <h1 id="heading">
84         jQuery Selectors Example
85     </h1>
86
87     <div class="box">
88         This is a DIV element with class selector.
89     </div>
90
91     <p>
92         This is a paragraph element.
93     </p>
94
95     <input type="text" placeholder="Enter your name">
```

```
97     <br><br>
98
99     <button id="idBtn">
100         ID Selector
101     </button>
102
103     <button id="classBtn">
104         Class Selector
105     </button>
106
107     <button id="elementBtn">
108         Element Selector
109     </button>
110
111     <button id="attrBtn">
112         Attribute Selector
113     </button>
114
115     <button id="jsonBtn">
116         Show JSON Data
117     </button>
118
119     <br><br>
120
121     <div id="jsonData" class="box">
122     </div>
123
124 </body>
125 </html>
```

Output:-

jQuery Selectors Example

This is a DIV element with class selector.

This is a paragraph element.

Enter your name

ID Selector

Class Selector

Element Selector

Attribute Selector

Show JSON Data

jQuery Selectors Example

This is a DIV element with class selector.

This is a paragraph element.

Aarti

ID Selector

Class Selector

Element Selector

Attribute Selector

Show JSON Data

jQuery Selectors Example

This is a DIV element with class selector.

This is a paragraph element.

Aarti

ID Selector

Class Selector

Element Selector

Attribute Selector

Show JSON Data

jQuery Selectors Example

This is a DIV element with class selector.

This is a paragraph element.

Aarti

ID Selector

Class Selector

Element Selector

Attribute Selector

Show JSON Data

jQuery Selectors Example

This is a DIV element with class selector.

This is a paragraph element.

Aarti

ID Selector

Class Selector

Element Selector

Attribute Selector

Show JSON Data

jQuery Selectors Example

This is a DIV element with class selector.

This is a paragraph element.

Aarti

ID Selector

Class Selector

Element Selector

Attribute Selector

Show JSON Data

Name: Aman
Course: BCA
City: Delhi

PRACTICAL – 3

Ques: Write a jQuery function to test whether a date is a weekend.

Input:

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Weekend Checker</title>
5
6      <!-- jQuery Library -->
7      <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
8
9      <style>
10
11          body{
12              font-family: Arial, sans-serif;
13              text-align: center;
14              margin-top: 50px;
15          }
16
17          input, button{
18              padding: 10px;
19              font-size: 18px;
20              margin: 10px;
21          }
22
23          #result{
24              font-size: 22px;
25              color: darkblue;
26              margin-top: 20px;
27          }
28      </style>
29
30      <script>
31
32          $(document).ready(function(){
33
34              $("#checkBtn").click(function(){
```



```
37         // Get date from textbox
38         let inputDate = $("#date").val();
39
40         // Convert to Date object
41         let date = new Date(inputDate);
42
43         // Get day number
44         // Sunday = 0
45         // Saturday = 6
46         let day = date.getDay();
47
48         // Check weekend
49         if(day == 0 || day == 6)
50         {
51             $("#result").html(
52                 "It is a Weekend"
53             );
54         }
55
56         else
57         {
58             $("#result").html(
59                 "It is a Weekday"
60             );
61         }
62
63     });
64
65 });
66
67 </script>
```

```
69 </head>
70
71 <body>
72
73     <h2>Check Whether Date is Weekend</h2>
74
75     <input type="date" id="date">
76
77     <button id="checkBtn">
78         Check
79     </button>
80
81     <div id="result"></div>
82
83 </body>
84 </html>
```


Output:-**Check Whether Date is Weekend**



Check

Check Whether Date is Weekend



Check

It is a Weekday

Check Whether Date is Weekend



Check

It is a Weekend

PRACTICAL – 4

Ques: Demonstrate various events: blur, change, focus, click, dblClick, submit.

Input:-

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>jQuery Events Example</title>
5
6      <!-- jQuery Library -->
7      <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
8
9      <style>
10
11          body{
12              font-family: Arial, sans-serif;
13              margin: 40px;
14          }
15
16          input, select, button{
17              padding: 10px;
18              margin: 10px 0;
19              font-size: 16px;
20          }
21
22          #box{
23              width: 250px;
24              height: 100px;
25              background-color: lightgray;
26              text-align: center;
27              line-height: 100px;
28              margin-top: 20px;
29              font-weight: bold;
30          }
31
32      </style>
33
34      <script>
35
36          $(document).ready(function(){
```

```
38         // Focus Event
39         $("#name").focus(function(){
40
41             $(this).css("background-color", "lightblue");
42
43         });
44
45         // Blur Event
46         $("#name").blur(function(){
47
48             $(this).css("background-color", "white");
49
50             alert("Textbox lost focus");
51
52         });
53
54         // Change Event
55         $("#course").change(function(){
56
57             alert("Course changed to: " + $(this).val());
58
59         });
60
61         // Click Event
62         $("#clickBtn").click(function(){
63
64             $("#box").html("Button Clicked!");
65
66         });
67
68         // Double Click Event
69         $("#box").dblclick(function(){
```

```
71         $(this).css("background-color", "yellow");
72
73         $(this).html("Double Click Detected!");
74
75     });
76
77     // Submit Event
78     $("#myForm").submit(function(){
79
80         alert("Form Submitted");
81
82     });
83
84 });
85
86 </script>
87
88 </head>
89
90 <body>
91
92     <h2>jQuery Events Demonstration</h2>
93
94     <!-- Form -->
95     <form id="myForm">
96
97         <!-- Focus and Blur -->
98         <label>Enter Name:</label><br>
99
100         <input type="text" id="name">
```

```
102         <br><br>
103
104         <!-- Change Event -->
105         <label>Select Course:</label><br>
106
107         <select id="course">
108
109             <option>B.Sc</option>
110             <option>BCA</option>
111             <option>B.Tech</option>
112             <option>MCA</option>
113
114         </select>
115
116         <br><br>
117
118         <!-- Click Event -->
119         <button type="button" id="clickBtn">
120             Click Me
121         </button>
122
123         <br><br>
124
125         <!-- Double Click Event -->
126         <div id="box">
127             Double Click Here
128         </div>
129
130         <br><br>
131
132         <!-- Submit Event -->
133         <button type="submit">
134             Submit Form
```

```
135         </button>
136
137     </form>
138
139 </body>
140 </html>
```

Output:-

jQuery Events Demonstration

Enter Name:

Select Course:

Click Me

Double Click Here

Submit Form

jQuery E

Enter Name:

Select Course:

Click Me

Double Click Here

Submit Form

This page says

Textbox lost focus

OK

jQuery Events Demonstration

Enter Name:

Select Course:

Button Clicked!

jQuery Events Demonstration

Enter Name:

Select Course:

Double Click Detected!

jQuery E

Enter Name:

Kirti

Select Course:

B.Sc

Click Me

Double Click Detected!

Submit Form

This page says

Form Submitted

OK

PRACTICAL – 5

Ques: Write a jQuery code that demonstrates the mouseOver and mouseOut event.

- a. The web page should have a hyperlink labelled “It shows the mouse events” and linked to your college website.
- b. When the mouse is hovered over the link the background colour of webpage changes to green and link text is changed to “I’m green now”.
- c. When the mouse is taken away from the link the background colour of webpage changes to red and the link text changes to “I’m red now”.

Input:

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4    <title>Mouse Events Example</title>
5
6    <!-- jQuery Library -->
7    <script src="https://code.jquery.com/jquery-3.6.0.min.js"></script>
8
9    <style>
10
11      body{
12        text-align: center;
13        margin-top: 100px;
14        background-color: red;
15        font-family: Arial, sans-serif;
16      }
17
18      a{
19        font-size: 28px;
20        text-decoration: none;
21        color: white;
22        font-weight: bold;
23      }
24
25    </style>
```

```
27     <script>
28
29         $(document).ready(function(){
30
31             // mouseOver Event
32             $("#collegeLink").mouseover(function(){
33
34                 $("body").css(
35                     "background-color", "green"
36                 );
37
38                 $(this).text("I'm green now");
39
40             });
41
42             // mouseOut Event
43             $("#collegeLink").mouseout(function(){
44
45                 $("body").css(
46                     "background-color", "red"
47                 );
48
49                 $(this).text("I'm red now");
50
51             });
52
53         });
54
55     </script>
56
57 </head>
```

```
59 <body>
60
61     <a id="collegeLink"
62       href="https://www.du.ac.in"
63       target="_blank">
64
65         It shows the mouse events
66
67     </a>
68
69 </body>
70 </html>
```

Output:-

A solid red rectangle.

I'm red now

A solid green rectangle.

I'm green now